THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 4 · Term 3

Student edition

Outcomes: MA-4.5 decimals (multiplying and dividing simply), MA-4.6 measurement, MA-4.7 angles and perimeter, MA-4.8 bar graphs.

Lesson notes

MA-4.5 and MA-4.6 Decimals and measurement (Weeks 1–4)

Multiply decimals by 10 and 100 by moving digits; divide a one-decimal number by a 1-digit number (4.8 ÷ 4 = 1.2). Convert km–m, m–cm, kg–g, L–mL; add and subtract mixed units (4 m 35 cm + 2 m 80 cm = 7 m 15 cm).

MA-4.7 Angles and perimeter (Weeks 5–8)

An angle is formed where two rays meet. Measure with a protractor (centre on the vertex, zero line on one arm). Acute < 90°, right = 90°, obtuse between 90° and 180°. Perimeter of polygons by adding sides.

MA-4.8 Bar graphs (Weeks 9–10)

Read and draw bar and block graphs with a title, labelled axes and a scale; class survey of favourite sports.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Measuring angles** — Open the angle and read it on the protractor. Watch its name change.

Workbook pages

Page	Outcome	Activity
1–2	MA-4.5, MA-4.6	Decimal × 10 and 100; unit conversions; mixed-unit sums.
3–4	MA-4.7	Measure and classify 10 angles; perimeters.
5	MA-4.8	Draw a bar graph from a tally chart.

Question bank

#	Outcome	Question	Your answer
1	MA-4.5	2.6 × 10 =	0.26 / 2.60 / 26 / 260
2	MA-4.5	4.8 ÷ 4 =	0.12 / 1.2 / 12 / 1.02
3	MA-4.6	3 km 250 m = ___ m	325 / 3,025 / 3,250 / 32,500
4	MA-4.6	2.5 kg = ___ g	250 / 2,500 / 25 / 25,000

#	Outcome	Question	Your answer
5	MA-4.6	1,750 mL = ___ L	17.5 / 1.75 / 175 / 0.175
6	MA-4.6	4 m 35 cm + 2 m 80 cm =	6 m 15 cm / 7 m 15 cm / 6 m 115 cm / 7 m 5 cm
7	MA-4.7	An angle of 90° is …	acute / right / obtuse / straight
8	MA-4.7	An angle of 130° is …	acute / right / obtuse / reflex
9	MA-4.7	An acute angle is less than 90°. True or false.	True / False
10	MA-4.7	Perimeter of a rectangle 12 cm by 7 cm =	19 cm / 84 cm / 38 cm / 26 cm
11	MA-4.8	[bar graph: cricket 12, football 8, hockey 4] How many more chose cricket than hockey?	_____
12	MA-4.8	Every bar graph needs a title and labelled axes. True or false.	True / False